COMPUTER PROGRAMMING · PRACTICE

Equal Sharing

DESCRIPTION

A teacher has n pencils and wants to hand them out to k students. Every student must receive the same number of pencils, and the teacher wants each student to receive as many as possible. Pencils that cannot be handed out evenly stay with the teacher.

Given n and k, find how many pencils each student receives and how many pencils are left over.

A function can return only one value, so the two answers must be passed back through pointers.

REQUIRED FUNCTION

Your program must define the following function:

```
void share(int n, int k, int *each, int *rest)
```

- `n` and `k` are normal input values (passed by value). The function stores the number of pencils each student receives into `*each`, and the number of pencils left over into `*rest`.
- It must store the answers through `*each` and `*rest`. It returns nothing.
- In `main`, read n and k, declare two more variables for the answers, call the function by passing their **addresses** with `&`, and then print the two variables.

INPUT FORMAT

The input contains one line with two integers n and k, separated by a single space.

$0 \le n \le 10^9$ and $1 \le k \le 10^9$. Note that n can be smaller than k, and n can be 0.

OUTPUT FORMAT

Print two integers on one line, separated by a single space: first the number of pencils each student receives, then the number of pencils left over. End the line with a newline.

SAMPLE

Sample 1 Input

```
17 5
```

Sample 1 Output

```
3 2
```